3 The Simplex Method for Solving linear Programming Problems

3.1 Elementary Matrix Operations

An elementary operation is either elementary row operation or elementary column operation and it does not change the original nature of the matrix.
These elementary operations are of three types.

1. The **interchange** of any two rows (or columns of a matrix).
   Notations: $Ri \leftrightarrow Rj$ or $Ci \leftrightarrow Cj$
   For example. Consider matrix **A** where

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

   We can interchange rows and columns in **A** as follows:

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} R1 \leftrightarrow R2 \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} \quad \text{Interchange row 1 and row 2}$$

   and

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} C2 \leftrightarrow C3 \begin{bmatrix} 0 & 2 & 1 \\ 1 & 3 & 2 \\ 3 & 1 & 1 \end{bmatrix} \quad \text{Interchange column 2 and column 3}$$

2. The multiplication of any row (or column) by a non-zero number.
   Notation: $kRi \to Ri$ or $kCi \to Ci$
   For example,

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} 3R1 \to R1 \begin{bmatrix} 0 & 3 & 6 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}; \quad \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} 2C3 \to C3 \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & 6 \\ 3 & 1 & 2 \end{bmatrix}$$

3. The addition of a multiple of one row (or column) to another row (or column).
   Notation: $kRj + Ri \to Ri$ or $kCj + Ci \to Ci$

   For example,

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} R1 + R2 \to R1 \begin{bmatrix} 1 & 3 & 5 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}; \quad \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} 2C3 + C1 \to C1 \begin{bmatrix} 4 & 1 & 2 \\ 7 & 2 & 3 \\ 5 & 1 & 1 \end{bmatrix}$$

3.2 The Simplex Method

The Simplex method is an approach to solving linear programming models using slack variables, tableaus, and pivot variables as a means to finding the optimal solution of an optimization problem. To solve a linear programming model using the Simplex method the following steps are necessary:

1. Writing the LPP in Standard form
2. Introducing slack variables
3. Creating the tableau
4. Identifying Pivot variables
5. Creating a new tableau
6. Checking for optimality
7. Identify optimal values

Step 1: Writing the LPP in Standard form

- Standard form is the baseline format for all linear programs before solving for the optimal solution and has three requirements: (1) must be a maximization problem, (2) all linear constraints must be in a less-than-or-equal-to ($\leq$) inequality, (3) all variables are non-negative ($x_i \geq 0$).
- These requirements can always be satisfied by transforming any given linear program using basic algebra and substitution.
- Standard form is necessary because it creates an ideal starting point for solving the Simplex method as efficiently as possible as well as other methods of solving optimization problems.
- To transform a minimization linear program model into a maximization linear program model, simply multiply both the left and the right sides of the objective function by -1.
- Transforming linear constraints from a greater-than-or-equal-to ($\geq$) inequality to a less-than-or-equal-to ($\leq$) inequality can be done similarly, by multiplying by -1 on both sides.
- Example of an LPP in standard form:

$$
\begin{aligned}
Maximize,\ z &= 8x_1 + 10x_2 + 7x_3 \\
subject \quad to &: \\
x_1 + 3x_2 + 2x_3 &\leq 10 \\
x_1 + 5x_2 + x_3 &\leq 8 \\
x_1, x_2, x_3 &\geq 0
\end{aligned}
$$

- Once the model is in standard form, the slack variables can be added as shown in Step 2 of the Simplex method.

Step 2: Introducing Slack Variables

- Slack variables are additional variables that are introduced into the linear constraints of a linear program to transform them from inequality constraints to equality constraints.
- If the model is in standard form, the slack variables will always have a +1 coefficient.
- Slack variables are needed in the constraints to transform them into solvable equalities with one definite answer.

$$\begin{aligned} x_1 + 3x_2 + 2x_3 + s_1 &= 10 \\ x_1 + 5x_2 + x_3 + s_2 &= 8 \\ x_1, x_2, x_3, s_1, s_2 &\geq 0 \end{aligned}$$

- After the slack variables are introduced, the tableau can be set up to check for optimality as described in Step 3.

Step 3: : Setting up the Tableau

- A Simplex tableau is used to perform row operations on the linear programming model as well as to check a solution for optimality.
- The tableau consists of the **coefficients** corresponding to the linear constraint variables and the coefficients of the objective function.
- The **top row** of the tableau states what each column represents. The **middle rows** represent the linear constraint variable coefficients from the linear programming model, and the **last row** represents the objective function variable coefficients

$$\begin{aligned} x_1 + 3x_2 + 2x_3 + s_1 &= 10 \\ x_1 + 5x_2 + x_3 + s_2 &= 8 \\ -8x_1 - 10x_2 - 7x_3 + z &= 0 \end{aligned}$$

x_1	x_2	x_3	s_1	s_2	z	b
1	3	2	1	0	0	10
1	5	1	0	1	0	8
-8	-10	-7	0	0	1	0

- Once the tableau has been completed, the model can be checked for an optimal solution as shown in Step 4.

Step 4: Checking Optimality

- The optimal solution of a maximization linear programming model are the values assigned to the variables in the objective function to give the largest zeta (z) value.
- To check optimality using the tableau, all values in the **last row** must contain values greater than or equal to zero (≥ 0). If a value is less than zero, it means that variable has not reached its optimal value.

- If a tableau is not optimal, the next step is to identify the pivot variable to base a new tableau on, as described in Step 5.

Step 5: Identifying Pivot variable

- The pivot variable is used in row operations to identify which variable will become the unit value and is a key factor in the conversion of the unit value.
- The pivot variable can be identified by looking at the **bottom row** of the tableau and the indicator.
- Assuming that the solution is not optimal, pick the **smallest negative value in the bottom row**. One of the values lying in the column of this value will be the pivot variable.
- To find the indicator, divide the (b) values of the linear constraints by their corresponding values from the column containing the possible pivot variable. The intersection of the row with the smallest non-negative indicator and the smallest negative value in the bottom row will become the **pivot variable/element**.
- Upon identifying the pivot variable, a new tableau is then created in Step 6 to optimize the variable and find the new possible optimal solution.

x1	x2	x3	s1	s2	z	b	Indicator
1	3	2	1	0	0	10	10/3
1	5	1	0	1	0	8	8/5
-8	-10	-7	0	0	1	0	

Smallest Value

Step 6: Creating a new tableau

- The new tableau will be used to identify a new possible optimal solution. **Row operations** can be performed to optimize the pivot variable while keeping the rest of the tableau equivalent.
- Once the new tableau has been completed, the model can be checked for an optimal solution.

x1	x2	x3	s1	s2	z	b
	0					
1/5	1	1/5	0	1/5	0	8/5
	0					

Pivot row

Pivot Column

Old Tableau:

x1	x2	x3	s1	s2	z	b
1	3	2	1	0	0	10
1	5	1	0	1	0	8
-8	-10	-7	0	0	1	0

Old pivot column

New Tableau:

x1	x2	x3	s1	s2	z	b
2/5	0	7/5	1	-3/5	0	26/5
1/5	1	1/5	0	1/5	0	8/5
-6	0	-5	0	2	1	16

New pivot row

Step 7: Checking Optimality

- As explained in Step 4, the optimal solution of a maximization linear programming model are the values assigned to the variables in the objective function to give the largest zeta value.
- Optimality will need to be checked after each new tableau to see if a new pivot variable needs to be identified.
- A solution is considered optimal if all values in the bottom row are greater than or equal to zero. If negative values exist, the solution is still not optimal and a new **pivot point** will need to be determined.

Step 8: Identifying optimal values

- Once the tableau is proven optimal the optimal values can be identified. These can be found by distinguishing the basic and non-basic variables.
- A **basic variable** can be classified to have a single 1 value in its column and the rest be all zeros. If a variable does not meet this criteria, it is considered **non-basic**.
- If a variable is non-basic it means the optimal solution of that variable is **zero**.
- If a variable is basic, the row that contains the 1 value will correspond to the *b* value. The *b* value will represent the optimal solution for the given variable.

x1	x2	x3	s1	s2	z	b
0	-2	1	1	-1	0	2
1	5	1	0	1	0	8
0	30	1	0	8	1	64

- Basic variables: x_1, s_1, z Non-basic variables: x_2, x_3, s_2
- For the variable x_1, the 1 is found in the second row. This shows that the optimal x_1 value is found in the second row of the b values, which is 8.
- Variable s_1 has a 1 value in the first row, showing the optimal value to be 2 from the b column. Due to s_1 being a slack variable, it is not actually included in the optimal solution since the variable is not contained in the objective function.
- The zeta variable has a 1 in the last row. This shows that the maximum objective value will be 64 from the b column.
- The final solution shows each of the variables having values of:

$$x_1 = 8; x_2 = 0; x_3 = 0; \qquad s_1 = 2; s_2 = 0; z = 64$$

- The maximum (optimal) value is 64 and found at $(8, 0, 0)$ of the objective function.

Example 3.2.1. *Solve the following LPP using the simplex method.*

$$\begin{aligned} &Maximize,\ p = 5x + 4y \\ &subject \quad to: \\ &\quad 3x + 5y \leq 78 \\ &\quad 4x + y \leq 36 \\ &\quad x, y \geq 0 \end{aligned}$$

Solution:

step 1: Writing the LPP in Standard form
The given LPP is already in standard form, i.e., (1) its a maximization problem, (2) all linear constraints have a less-than-or-equal-to ($\leq$) inequality and (3) all variables are non-negative ($x, y \geq 0$).

step 2: Introducing slack variables
Let u and w be our slack variables. We insert the slack variables to turn the inequalities into equations. Ensure p (to be maximized) is positive and appears in the last equation.

$$\begin{aligned} 3x + 5y + u &= 78 \\ 4x + y + w &= 36 \\ -5x - 4y + p &= 0 \end{aligned}$$

step 3: Creating the tableau Each row is the coefficient of each of the equations above.

x	y	u	w	p	b
3	5	1	0	0	78
4	1	0	1	0	36
-5	-4	0	0	1	0

step 4: Identifying Pivot element/value

From the botton row, identify the smallest negative number. In this case, its -5 in column 1. So column 1 is the **pivot column**. Next, divide the b values in the last column by corresponding values in the pivot column 1.

x	y	u	w	p	b	
3	5	1	0	0	78	26
$\boxed{4}$	1	0	1	0	36	9
-5	-4	0	0	1	0	

Since 9 is smallest, row 2 is the **pivot row** and 4 is the **pivot element**.

step 5: Creating a new tableau

Use row operations to reduce the pivot element 4 to 1.
Perform: $\frac{1}{4}R2 \to R2$.

x	y	u	w	p	b
3	5	1	0	0	78
1	1/4	0	1/4	0	9
-5	-4	0	0	1	0

Next, reduce the rest of the values in the pivot column to zero. Perform the operations: $3R2 - R1 \to R1$ and $5R2 + R3 \to R3$.

x	y	u	w	p	b
0	17/4	1	-3/4	0	51
1	1/4	0	1/4	0	9
0	-11/4	0	5/4	1	45

Since we still have a negative element in the bottom row, we repeat step 4 and identify a new pivot element. The negative element (-11/4) occurs in the second column. Therefore column 2 is the new pivot column. Upon diving the values under column b with corresponding pivot column values, we have

x	y	u	w	p	b	
0	$\boxed{17/4}$	1	-3/4	0	51	12
1	1/4	0	1/4	0	9	36
0	-11/4	0	5/4	1	45	

Since 12 is the smallest value, row 1 is our new pivot row and 17/4 is the new pivot element. Next, reduce the pivot element to 1 using the operation: $\frac{4}{17}R1 \to R1$.

x	y	u	w	p	b
0	1	4/17	-3/17	0	12
1	1/4	0	1/4	0	9
0	-11/4	0	5/4	1	45

Next, get zeros (0's) elsewhere in the pivot column using the operations: $\frac{1}{4}R1 - R2 \to R2$ and $\frac{11}{4}R1 + R3 \to R3$.

x	y	u	w	p	b
0	1	4/17	-3/17	0	12
1	0	-1/17	7/17	0	6
0	0	11/17	13/17	1	78

step 6: Checking for optimality

A solution is considered optimal if all values in the bottom row are greater than or equal to zero. Since all the values in the bottom row are all non-negative (0, 0, 11/17, 52/51, 1), we have an optimal solution. Stop and read the solution.

step 7: Identify optimal values

- From step 6 above, the **Basic variables** are x, y and p.
- The **Non-basic variables** are u and w.
- For the variable x, the 1 is found in the second row. This shows that the optimal x value is found in the second row of the b values, which is 12.
- For the variable y, the 1 is found in the first row. This shows that the optimal y value is found in the first row of the b values, which is 6.
- The p variable has a 1 in the last row. This shows that the maximum objective value will be 78 from the b column.
- The final solution shows each of the variables having values of:

$$x = 12; y = 6; \qquad u = 0; w = 0; \qquad p = 78$$

- The maximum (optimal) value of 78 occurs at $(12, 6)$ of the objective function.

Example 3.2.2. *Solve the following LPP using the simplex method.*

$$\begin{aligned} Maximize,\ p &= 3x + 4y + z \\ subject\quad to &: \\ 3x + 10y + 5z &\leq 120 \\ 5x + 2y + 8z &\leq 6 \\ 8x + 10y + 3z &\leq 105 \\ x, y, z &\geq 0 \end{aligned}$$

Solution:
The LPP is already in standard form. Introducing slack variables u, v and w yields:

$$\begin{aligned} 3x + 10y + 5z + v &= 120 \\ 5x + 2y + 8z + u &= 6 \\ 8x + 10y + 3z + w &= 105 \\ -3x - 4y - z + p &= 0 \end{aligned}$$

The initial tableau is given by

x	y	z	v	u	w	p	b
3	10	5	1	0	0	0	120
5	2	8	0	1	0	0	6
8	10	3	0	0	1	0	105
-3	-4	-1	0	0	0	1	0

The second column has the smallest negative value (-4). Upon diving the values in column b with the corresponding values in the pivot column 2, we obtain

x	y	z	v	u	w	p	b	
3	10	5	1	0	0	0	120	12
5	$\boxed{2}$	8	0	1	0	0	6	3
8	10	3	0	0	1	0	105	10.5
-3	-4	-1	0	0	0	1	0	

Hence, the pivot element is 2 (row 2, column 2).
Using the row operation $\frac{1}{2}R2 \rightarrow R2$, we reduce the pivot element 2 to 1.

x	y	z	v	u	w	p	b
3	10	5	1	0	0	0	120
5/2	1	4	0	1/2	0	0	3
8	10	3	0	0	1	0	105
-3	-4	-1	0	0	0	1	0

Next, reduce the other elements/values in the pivot column to 0. We perform the following row operations:

$$\begin{aligned} -10R2 + R1 &\rightarrow R1 \\ -10R2 + R3 &\rightarrow R3 \\ 4R2 + R4 &\rightarrow R4 \end{aligned}$$

This yields:

x	y	z	v	u	w	p	b
-22	0	-35	1	-5	0	0	90
5/2	1	4	0	1/2	0	0	3
-17	0	-37	0	-5	1	0	75
7	0	15	0	2	0	1	12

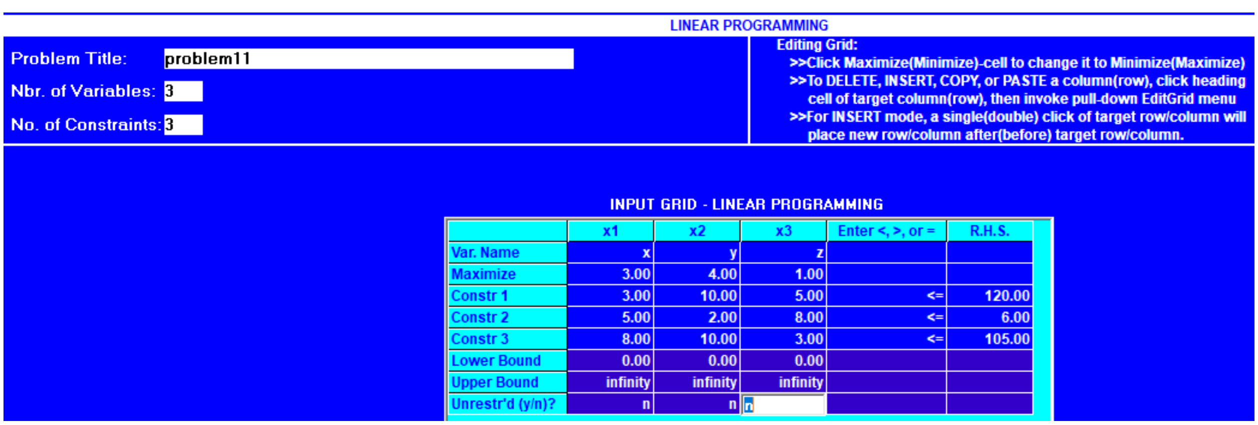
Notice that there are no more negative values/numbers in the bottom row. We therefore stop and read the optimal values.
We can clearly see that the **Basic variables** are y, v, w and p (They have a 1 and 0's only). The **Non-basic variables** are x, z and u (do not have a 1 and 0's only).
The final solution is:

$$x = 0; y = 3; z = 0; v = 90; u = 0; w = 75; p = 12$$

The above optimal solution can also be obtained using the **Simplex (Algebraic) method** in TORA software as follows:

TORA moment
Main Menu $\Longrightarrow$ Linear Programming $\Longrightarrow$ Enter required data $\Longrightarrow$ Solve Problem $\Longrightarrow$ Algebraic $\Longrightarrow$ Iterations $\Longrightarrow$ All-slack starting solution (for Max. problem only).

The input data is:



	x1	x2	x3	Enter <, >, or =	R.H.S.
Var. Name	x	y	z		
Maximize	3.00	4.00	1.00		
Constr 1	3.00	10.00	5.00	<=	120.00
Constr 2	5.00	2.00	8.00	<=	6.00
Constr 3	8.00	10.00	3.00	<=	105.00
Lower Bound	0.00	0.00	0.00		
Upper Bound	infinity	infinity	infinity		
Unrestr'd (y/n)?	n	n	n		

The results are:

Next Iteration | All Iterations | Write to Printer

Iteration 1	x	y	z				
Basic	x1	x2	x3	sx4	sx5	sx6	Solution
z (max)	-3.00	-4.00	-1.00	0.00	0.00	0.00	0.00
sx4	3.00	10.00	5.00	1.00	0.00	0.00	120.00
sx5	5.00	2.00	8.00	0.00	1.00	0.00	6.00
sx6	8.00	10.00	3.00	0.00	0.00	1.00	105.00
Lower Bound	0.00	0.00	0.00				
Upper Bound	infinity	infinity	infinity				
Unrestr'd (y/n)?	n	n	n				
Iteration 2	x	y	z				
Basic	x1	x2	x3	sx4	sx5	sx6	Solution
z (max)	7.00	0.00	15.00	0.00	2.00	0.00	12.00
sx4	-22.00	0.00	-35.00	1.00	-5.00	0.00	90.00
x2	2.50	1.00	4.00	0.00	0.50	0.00	3.00
sx6	-17.00	0.00	-37.00	0.00	-5.00	1.00	75.00
Lower Bound	0.00	0.00	0.00				
Upper Bound	infinity	infinity	infinity				
Unrestr'd (y/n)?	n	n	n				

The optimal solution is:

$$x = 0; y = 3; z = 0; v = 90; u = 0; w = 75; p = 12$$

For minimization problem with two-phases, follow the steps below:

TORA moment
Main Menu $\Longrightarrow$ Linear Programming $\Longrightarrow$ Enter required data $\Longrightarrow$ Solve Problem $\Longrightarrow$ Algebraic $\Longrightarrow$ Iterations $\Longrightarrow$ Two-phase method (for Min. problem). Results.

Exercise 2.

1. Solve the above Reddy Mikks model (in Example 1) using the simplex method. The LPP model is:

$$\begin{aligned} &\text{Maximize } z = 5x_1 + 4x_2 \\ &\text{subject to:} \\ &\qquad 6x_1 + 4x_2 \leq 24 \\ &\qquad x_1 + 2x_2 \leq 6 \\ &\qquad -x_1 + x_2 \leq 1 \\ &\qquad x_2 \leq 2 \\ &\qquad x_1, x_2 \geq 0 \end{aligned}$$

   Solution: $x_1 = 3; x_2 = 1.5; z = 21,000$

2. Solve using the Simplex method the following problem:

$$\begin{aligned} \text{Maximize } z = f(x,y) = 3x + 2y& \\ \text{subject to :} 2x + y &\leq 18 \\ 2x + 3y &\leq 42 \\ 3x + y &\leq 24 \\ x \geq 0; y &\geq 0 \end{aligned}$$

   Solution: $x = 3; y = 12; z = 33$

3. Find the minimum value of $w = 0.12x_1 + 0.15x_2$ subject to the following constraints:

$$\begin{aligned} 60x_1 + 60x_2 &\geq 300 \\ 12x_1 + 6x_2 &\geq 36 \\ 10x_1 + 30x_2 &\geq 90 \\ x_1 \geq 0; x_2 &\geq 0 \end{aligned}$$

Solution: $x_1 = 3; x_2 = 2; z = 33/50$

4. Find the solution to the minimization problem:

$$\begin{aligned} \text{Minimiz } z &= 12x_1 + 16x_2 \\ \text{subject to :} x_1 + 2x_2 &\geq 40 \\ x_1 + x_2 &\geq 30 \\ x_1 \geq 0; x_2 &\geq 0 \end{aligned}$$

Solution: ${}_1x = 20; x_2 = 10; z = 400$

3.3 LP Models: Investment Problem

In finance, financial decisions are taken based on the trench hold between risk and return. Investors are usually faced with problems of what asset to invest into and how to come up with an investment portfolio that can provide the highest return and also lowest risk level. Linear programming has been used to assist investors to come up with the best investment portfolio.

To illustrate this, consider the following problem.

Example 3.3.1.

Suppose you have \$12,000 to invest, and three different funds from which to choose: the municipal bonds fund (with a 7% return), the local bank's CDs (with an 8% return) and the high-risk account (with a 12% return). To minimize the risk, you decide not to invest any more than \$2000 in the high-risk account. For tax reasons, you need to invest at least three times as much in the municipal bonds as in the bank CDs. Assuming the year end yields are as expected, what are the optimal investment amounts.

Part A: LP model formulation:
Step 1: Define the decision variables. Let

x_1: amount invested in bonds.

x_2: amount invested in bank CDs.

x_3: amount invested in high-risk accounts

Step 2: Write the objective function
The objective is to maximize interest revenue. Therefore,

$$\text{Maximize} \quad z = 0.07x_1 + 0.08x_2 + 0.12x_3$$

Step 3: Write the constraints

Total available: $x_1 + x_2 + x_3 \leq 12000$

high-risk: $x_3 \leq 2000$

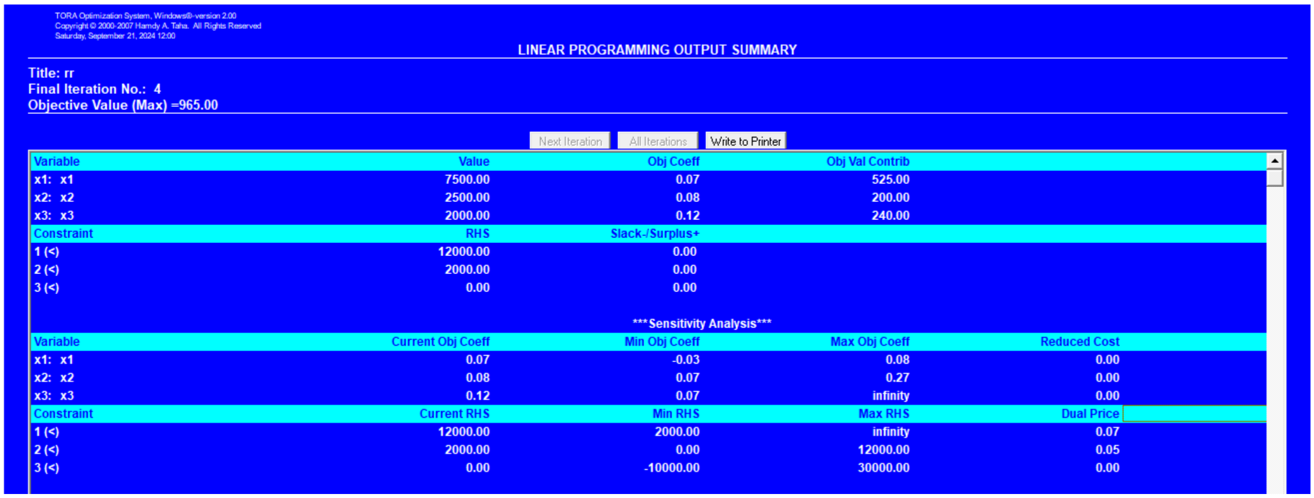
tax requirements: $x_1 \geq 3x_2$

Trivial/non-negativity: $x_1 \geq 0$, $x_2 \geq 0$, $x_3 \geq 0$

The final LP model is:

$$\begin{aligned} \text{Maximize } z = 0.07x_1 + 0.08x_2 + 0.12x_3 & \\ \text{subject to:} \quad x_1 + x_2 + x_3 &\leq 12000 \\ x_3 &\leq 2000 \\ -x_1 + 3x_2 &\leq 0 \\ x_1 \geq 0, x_2 \geq 0, x_3 &\geq 0. \end{aligned}$$

Part B: LP model Solution:
Using the Simplex Algorithm in TORA software, the solution to the above LP program is:
Invest \$7500 in municipal bonds, \$2500 in bank CDs and \$2000 in the high-risk account; for the maximum return or revenue or profit of \$965.



TORA Optimization System, Windows®-version 2.00
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Saturday, September 21, 2024 12:00

LINEAR PROGRAMMING OUTPUT SUMMARY

Title: rr
Final Iteration No.: 4
Objective Value (Max) =965.00

Next Iteration | All Iterations | Write to Printer

Variable	Value	Obj Coeff	Obj Val Contrib
x1: x1	7500.00	0.07	525.00
x2: x2	2500.00	0.08	200.00
x3: x3	2000.00	0.12	240.00

Constraint	RHS	Slack-/Surplus+
1 (<)	12000.00	0.00
2 (<)	2000.00	0.00
3 (<)	0.00	0.00

Sensitivity Analysis

Variable	Current Obj Coeff	Min Obj Coeff	Max Obj Coeff	Reduced Cost
x1: x1	0.07	-0.03	0.08	0.00
x2: x2	0.08	0.07	0.27	0.00
x3: x3	0.12	0.07	infinity	0.00

Constraint	Current RHS	Min RHS	Max RHS	Dual Price
1 (<)	12000.00	2000.00	infinity	0.07
2 (<)	2000.00	0.00	12000.00	0.05
3 (<)	0.00	-10000.00	30000.00	0.00

3.4 LP Models: Asset/Liability Cash Flow Matching (Short term financing)

Corporations routinely face the problem of financing short term cash commitments. Linear programming can help in figuring out an optimal combination of financial instruments to meet these commitments. To illustrate this, consider the following problem. For simplicity of exposition, we keep the example very small.

A company has the following short term financing problem.

Month	Jan	Feb	Mar	Apr	May	Jun
Net Cash Flow	-150	-100	200	-200	50	300

Net cash ow requirements are given in thousands of dollars. The company has the following sources of funds

- A line of credit of up to \$100K at an interest rate of 1% per month,
- In any one of the first three months, it can issue 90-day commercial paper bearing a total interest of 2% for the 3-month period,
- Excess funds can be invested at an interest rate of 0.3

There are many questions that the company might want to answer: What interest payments will the company need to make between January and June? Is it economical to use the line of credit in some of the months? If so, when? How much?

Linear programming gives us a mechanism for answering these questions quickly and easily. It also allows to answer some "what if" questions about changes in the data without having to resolve the problem. What if Net Cash Flow in January were -200 (instead of -150)? What if the limit on the credit line were increased from 100 to 200? What if the negative Net Cash Flow in January is due to the purchase of a machine worth 150 and the vendor allows part or all of the payment on this machine to be made in June at an interest of 3% for the 5-month period? The answers to these questions are readily available when this problem is formulated and solved as a linear program.

A) Modelling: LP Model formulation:

- We begin by modelling the above short term financing problem. That is, we write it in the language of linear programming. There are rules about what one can and cannot do within linear programming. These rules are in place to make certain that the remaining steps of the process (solving and interpreting) can be successful.

- Key to a linear program are the *decision variables*, *objective*, and *constraints.*

- During the 6 months, we should be funding short term capital to fulfill the cash flow requirements. Given that the sources of funds have different interests and terms, we can set up a mathematical optimization problem for this work. We will use the company's wealth v in June as the objective function and the following decision variables:

  x_i: the line of credit in month $i (i = 1, 2, 3, 4, 5)$

  y_i: commercial paper issued in month $i (i = 1, 2, 3)$

  z_i: excess funds in month $i (i = 1, 2, 3, 4, 5)$

  $v = z_6$: company's wealth in June.

- In this case, the objective function is simply to *maximize v.*

- In January ($i = 1$), there is a cash requirement of \$150. To meet this requirement, the company can draw an amount x_1 from its line of credit and issue an amount y_1 of commercial paper. Considering the possibility of excess funds z_1, the cash flow balance equation is as follows.

$$x_1 + y_1 - z_1 = 150.$$

- Next, in February ($i = 2$), there is a cash requirement of \$100. In addition, principal plus interest of $1.01x_1$ is due on the line of credit and $1.003z_1$ is received on the invested excess funds. To meet the requirement in February, the company can draw an amount x_2 from its line of credit and issue an amount y_2 of commercial paper. So, the cash flow balance equation for February is as follows.

$$x_2 + y_2 - 1.01x_1 + 1.003z_1 - z_2 = 100.$$

- Similarly in March, we get the following equation.

$$x_3 + y_3 - 1.01x_2 + 1.003z_2 - z_3 = -200.$$

- For the months of April, May, and June, issuing a commercial paper is no longer an option, so we will not have variables y_4, y_5 and y_6 in the formulation. Furthermore, any commercial paper issued between January and March requires a payment with 2% interest 3 months later. Thus, we have the following additional equations:

$$\begin{aligned} x_4 - 1.02y_1 - 1.01x_3 + 1.003z_3 - z_4 &= 200 \\ x_5 - 1.02y_2 - 1.01x_4 + 1.003z_4 - z_5 &= -50 \\ - 1.02y_3 - 1.01x_5 + 1.003z_5 - v &= -300. \end{aligned}$$

Note that x_i is the balance on the credit line in month i, not the incremental borrowing in month i. Similarly, z_i represents the overall excess funds in month i. This choice of variables is quite convenient when it comes to writing down the upper bound and non-negativity constraints.

$$\begin{aligned} 0 \leq x_i &\leq 100 \\ y_i &\geq 0 \\ z_i &\geq 0. \end{aligned}$$

Thus, the complete LP model of this problem is:

$$\begin{aligned} \text{Maximize } & v \\ \text{subject to: } & x_1 + y_1 - z_1 = 150 \\ & x_2 + y_2 - 1.01x_1 + 1.003z_1 - z_2 = 100 \\ & x_3 + y_3 - 1.01x_2 + 1.003z_2 - z_3 = -200 \\ & x_4 - 1.02y_1 - 1.01x_3 + 1.003z_3 - z_4 = 200 \\ & x_5 - 1.02y_2 - 1.01x_4 + 1.003z_4 - z_5 = -50 \\ & - 1.02y_3 - 1.01x_5 + 1.003z_5 - v = -300. \\ & x_1 \leq 100 \\ & x_2 \leq 100 \\ & x_3 \leq 100 \\ & x_4 \leq 100 \\ & x_5 \leq 100 \\ & x_i, y_i, z_i \geq 0 \end{aligned}$$

Formulating a problem as a linear program means going through the above process of clearly defining the decision variables, objective, and constraints.

B) Solving the LP Model

Special computer programs can be used to find solutions to linear programming models. The most widely available program is undoubtedly SOLVER, included in all recent versions of the Excel spreadsheet program. other suggestions include: CLP, TORA, MATLAB, AMPL etc.

Using the solver in TORA, the company's wealth v in June will be \$92,497. To achieve this, the company will issue \$150,000 in commercial paper in January, \$49,020 in February and \$203,434 in March. In addition, it will draw \$50,980 from its line of credit in February. Excess cash of \$351,944 in March will be invested for just one month.

Exercise 3. *How would the formulation of the short-term financing problem above change if the commercial papers issued had a 2 month maturity instead of 3?*

Exercise 4. *A company will face the following cash requirements in the next eight quarters (positive entries represent cash needs while negative entries represent cash surpluses).*

Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8
100	500	100	-600	-500	200	600	-900

The company has three borrowing possibilities.

- *a 2-year loan available at the beginning of Q_1 , with a 1% interest per quarter.*
- *The other two borrowing opportunities are available at the beginning of every quarter: a 6-month loan with a 1.8% interest per quarter, and a quarterly loan with a 2.5% interest for the quarter.*

Any surplus can be invested at a 0.5% interest per quarter. Formulate a linear program that maximizes the wealth of the company at the beginning of Q_9.

Exercise 5. *A home buyer in France can combine several mortgage loans to finance the purchase of a house. Given borrowing needs B and a horizon of T months for paying back the loans, the home buyer would like to minimize his total cost (or equivalently, the monthly payment p made during each of the next T months). Regulations impose limits on the amount that can be borrowed from certain sources. There are n different loan opportunities available. Loan i has a fixed interest rate r_i, a length $T_i \leq T$ and a maximum amount borrowed b_i. The monthly payment on loan i is not required to be the same every month, but a minimum payment m_i is required each month. However the total monthly payment p over all loans is constant. Formulate a linear program that finds a combination of loans that minimizes the home buyer's cost of borrowing. [Hint: In addition to variables x_{ti} for the payment on loan i in month t, it may be useful to introduce a variable for the amount of outstanding principal on loan i in month t.]*